

Ch1 选择

1. The two principal **catalysts** for the **Information Age** have been

=> computers and communication networks. <

Catalysts:

- Low-cost computers
- High-speed communication network

2. Which statement best supports the conclusion that **society** can **control** whether to adopt a new technology?

b) Despite decades of research, fusion power is an elusive goal. 核聚变发电是一个难以实现的目标

e) **After** the accident at Three Mile Island, 67 nuclear reactor construction projects **were canceled**. <

3. Tablets 计算板, abacuses 算盘, and manual tables 手写表

Aids to Manual Calculating (1 of 2)

- Tablet to record numbers
 - Clay, **wax tablets** (ancient times)
 - Slates (late Middle Ages)
 - Paper tablets (19th century to present)
- **Abacus** to perform arithmetic
 - Rods or wires in rectangular frame
 - Lines drawn on a counting board



a) are no longer used, because of the proliferation of calculators and computers.

b) **are examples of aids 辅助物 to manual calculating.** <

c) were developed in Western Europe in the late Middle Ages.

算盘的历史可以追溯到古代中国、罗马和中东，而不是晚期中世纪的西欧。

计算表格也并非仅限于西欧，其概念在多个文明中独立发展

d) replaced Hindu–Arabic numerals as the preferred way to do calculations. 这些只是计算辅助工具，没有取代数字系统

4. The mechanical adding machines of Pascal 帕斯卡 and Leibniz 莱布尼茨 were not widely adopted because

- Calculators of *Pascal* (French tax collector, 1640) and after *Leibniz* (German)
 - Worked with whole numbers
 - Unreliable

b) **there were unreliable.** < 这些机械计算器易损坏且计算不够稳定，尤其是在齿轮和机械部件方面容易出错或卡住

c) they were too difficult to program. 这些机械计算器主要用于加法和乘法，根本不涉及编程

d) they could not handle fractions 分数. 这些计算器主要用于整数运算，而分数运算通常在数学表格或人工计算中解决

5. The calculating machine of **Georg and Edvard Sheutz**

a) computed the values of polynomial functions. 多项式

b) typeset the results of its computations. 排版计算结果

c) faster.

d) more reliably

e) **All of the above.** <

6. Which of the following phrases does not describe the **Gilded Age** in the United States?
美国镀金时代

a) Rapid industrialization

b) Economic expansion

c) **Widespread electrification** < 电气化仍然局限于部分城市、未广泛普及

d) Concentration of corporate power

e) Corporate mergers

7. Which of the following was not a result of the adoption of mechanical calculators?

- a) 减少对于高计算能力者的需求
 - b) Higher productivity of bookkeepers 提升了记账员的生产力
 - c) Higher salaries of bookkeepers <
 - d) Proliferation of companies making calculators 促进了计算器制造企业的发展
 - e) Feminization of bookkeeping
 - Adopt mechanical (adding) calculator
 - “Deskilling” and feminization of bookkeeping
 - People of average ability quite productive
 - Calculators 6× faster than adding by hand
 - Wages dropped 工资下降
 - Women replaced men
-

8. Which of the following was not a feature of cash registers in the early 1900s?

- a) Ability to compute total of purchases
- b) Ability to print itemized receipts for customers
- c) Ability to print log of transactions for owners
- d) Ability to compute amount of **change to give customer** <
- e) Ability to ring a bell every time cash drawer is opened

Cash Register (1 of 2)

- In late 1800s smaller stores evolved into “department stores” with multiple departments
 - Two problems faced by department store owners
 - Keeping accurate sales records
 - Preventing fraud from clerks 职员
 - **Response to problems: Cash Register**
 - Created printed, itemized receipts 生成打印的、逐项列出的收据
 - Maintained printed log of transactions 保存交易的打印日志
 - Rang bell every time drawer was opened 每次打开抽屉时响铃
 - Become an important **information-processing device**
-

9. **Punched card tabulation** was invented by Herman Hollerith when he was an employee of

b) the Census 人口普查 Bureau 局 <

Tabulators → Data-Processing Systems

- Data-processing system
 - Receives input data
 - Performs one or more calculations
 - Produces output data
- Punched cards
 - Stored input data and intermediate results
 - Stored output
 - On most sophisticated systems, also stored **programs**

将打孔卡应用在数据处理系统中

10. Which of the following phrases best describes a machine that inputs data, performs one or more calculations, and produces output data?

c) **Data-processing system** <

11. The **first commercial electronic digital computers** were produced just after

c) **World War II.** <

12. Programming languages **FORTRAN** and **COBOL** became international standards, meaning

Programming Languages

- Assembly language
 - Symbolic representations of machine instructions
 - Programs just as long as machine language programs
- FORTRAN
 - First higher-level language (shorter programs)
 - Designed for scientific applications
 - Initially, people are skeptical about quality of compiler generated machine code
 - Programmers wrote 5-20x faster
- COBOL
 - U.S. Department of Defense standard
 - Designed for business applications

d) they were supported by every computer manufacturer. <

13. Software that allows **multiple users to edit and run** their programs simultaneously on the same computer is called

- Time-Sharing Systems
 - Divide computer time among multiple users
 - Users connect to computer via terminals
 - Cost of ownership spread among more people
 - Gave many more people access to computers

e) a time-sharing system. <

14. Which **Cold War program** played an important role in advancing **integrated circuit** technology? 冷战时期推动集成电路技术发展的项目

- a) B-52 bomber
 - b) Hydrogen bomb
 - c) Mark 37 torpedo
 - d) Minuteman II ballistic missile < 民兵 II 弹道导弹
 - e) NORAD radar network
-

15. Which company **produced the System/360**, a family of 19 compatible mainframe computers?

c) **IBM** <

16. The company that invented the **microprocessor** is

d) **Intel** <

First microprocessor – Intel 4004

17. A key **application** that **first made personal computers more attractive to business** was

a) the spreadsheet program. <

个人计算机对商业的影响:

1. VisiCalc (visible calculator)
 - a. spreadsheet program 电子制表软件
 - b. Apple II 的杀手软件 killer application
 - c. 影响: **Turning the microcomputer from a hobby for computer enthusiasts into a serious business tool.**
-

18. The software company that provided **IBM** with the operating system for its PC was

c) **Microsoft**. <

- **MS-DOS 操作系统**: 1980 年代初, IBM 计划推出个人电脑 (IBM PC) , 但需要一个操作系统。IBM 找到了 Microsoft, 最终采用了 **MS-DOS** 作为 IBM PC 的操作系统
-

19. The **first** electronic networking technology widely used in the **United States** was the

c) **telegraph**. < 电报 – 第一种电子通信网络技术

A machine that could send messages long distances across wires

Telegraph (1 of 2)

- U.S. government funded first line
 - 40 miles from Washington, D.C. to Baltimore
 - Built by Samuel Morse in 1843-1844
- Private networks flourished
 - 12,000 miles of lines in 1850
 - Transcontinental line in 1861 put Pony Express out of business
 - 200,000 miles of lines by 1877

20. The **Pony Express** went out of business when 驿马快递 (Pony Express) 为何停业?

c) the transcontinental **telegraph** 横贯大陆电报线 was completed. <

- 驿马快递 是 19 世纪美国西部的重要快递服务，依靠骑马传递邮件。
 - 1861 年横贯大陆电报建成，使得电报取代了邮递作为主要信息传输方式，导致驿马快递停业。
-

21. Alexander Graham Bell invented the **harmonic or musical telegraph**, which enabled

a) more than one **message** to be **sent over a single telegraph** wire at the same time. < 它可以让多个**信号**同时在一条电报线传输。

b) human speech to be sent over a telegraph wire.

c) music to be sent over a telegraph wire.

22. Most early telephones were **installed in businesses**, because

e) leasing 出租 a telephone was expensive. <

Social Impact of Telephone

- Blurred public life / private life boundary
 - Eroded traditional social hierarchies
 - Reduced privacy
-

23. A typewriter that **prints** a message transmitted over a **telegraph** line is called a 电报线路上传输信息的打印设备

c) **teletype**. < 电传打字机一种能够自动接收并打印电报信息的机械打字机。

24. The power of **radio** as a medium of mass communication 大众传播 was demonstrated in 1938, when Orson Welles put on a dramatization 改编成戏剧 of

a) **War of the Worlds**. <

当时奥森·威尔斯以新闻报道的风格播放了《世界大战》的广播剧，引发了部分听众的恐慌，他们误认为外星人真的入侵地球。这一事件证明了广播作为大众传播媒介的强大影响力。

25. ARPA Director **J.C.R. Licklider** conceived of a Galactic **Network** that would

c) become the world's most powerful number-crunching machine.

d) facilitate the exchange of programs and data. <

Licklider 设想了一种全球计算机网络，帮助科学家、工程师等共享数据和程序，这也是互联网理念的雏形。

26. One of the first and most important applications of the **ARPANET** was ARPANET 最早的重要应用

a) email. <

27. What term is used to describe a **high-speed Internet connection**? 什么术语描述高速互联网连接?

a) broadband < 宽带 (Broadband) 指 高速互联网连接，比拨号上网更快、更稳定

Broad-band 宽带

1. High-speed Internet connection 是一种高速互联网连接方式 与传统的拨号网络相比，速度快得多。
 2. Makes feasible transfer of very large files (如, video)
 3. 推动了file-swapping(文件共享)
-

28. The **utilization of remote computing resources** over the Internet is called

a) cloud computing. <

Cloud computing

1. 是什么? Utilization of remote computing resources over Internet
-

- Enablers of cloud computing
 - High-speed networks
 - High-performance, low-cost microprocessors
 - Virtualization: software allowing a single physical computer to emulate many virtual computing devices

29. Enabling a single physical computer to emulate 模仿 many virtual computing devices is called

d) virtualization. <

30. The first alphabet 字母表 to represent vowels 元音 as well as consonants 辅音 was developed by the

d) **Greeks.** <

Greek Alphabet

- True alphabet: letters for both consonant and vowel sounds
 - 750 BC: Greeks developed first true alphabet with 24 characters
 - Simple, efficient way of transforming spoken words into written form
-

31. In the fourth century the codex replaced the scroll because

书册取代卷轴？

a) it was more durable 耐用, and it was much easier to look up a particular passage. <

- Codex
 - Rectangular pages sewn together on one side
 - Replaced papyrus scrolls as way of storing books
 - Allowed quicker access to particular passages
 - First produced by hand, then by wood engraving
-

32. The technology to **mass-produce paper** was developed in

a) China. < 蔡伦改进造纸术（公元 105 年），推动了纸张的大规模生产。

33. **Hypertext** is supposed to mimic 消防

c) the associative memory of human beings. <

超文本的结构类似于人类的思维方式，能够通过链接建立关联。

34. What visionary 有远见的人 invented the computer mouse and demonstrated windows, email, and live network videoconferencing at "**the mother of all demos**" in 1968?
- b) Douglas Engelbart <
 - c) Bill Gates 微软
 - d) Steve Jobs Apple
-

35. The first popular personal computer with a **graphical user interface** was the
- a) **Apple Macintosh.** <

GUI Graphical User Interface

1. Xerox PARC (Palo Alto Research Center)
 - a. Alan Kay 在 1968年目睹了 Douglas Engelbart 的“所有演示之母” (The Mother of All Demos) , 深受启发, 开始研究图形用户界面。
 - b. Alto personal computer – early 1970s
 - i. 技术: 拥有Bit-mapped display位图显示器、keyboard、mouse
 - ii. 遗憾: 未能商业化成功
 2. Apple computer
 - a. 1979, Steve Jobs 参观了 Xerox PARC, 并看到了他们的图形用户界面技术。
 - b. Macintosh 1984 – 拥有Bit-mapped display位图显示器、keyboard、mouse
 3. Microsoft Windows 3.0 for IBM PCs
 - a. Released in May 1990
 - b. windows3.0是微软为IBM PC 推出的图形用户界面系统
 - c. Quickly became dominant graphical user interface
-

36. A program that **accepts a list of keywords** and **returns documents most closely matching** them is called a(n)
- b) object-oriented database.
 - c) relational database.
 - d) search engine. <
- Search engine
 - Accepts list of keywords from user
 - Searches a database of documents
 - Returns documents most closely matching keywords
-

37. **Storing data remotely** on **servers** accessed through the Internet is often called

a) cloud storage. <

(R3)

38. The Bell Telephone Company was able to establish a monopoly on telephone services in the United States because

a) Alexander Graham Bell held a **patent** on the telephone. <

- Alexander Graham Bell and Thomas A. Watson succeeded in transmitting speech electronically in 1876.
 - Installing a phone was expensive! After Bell's first patent expired, the home-use phones increased rapidly.
-